
WEEK_6:-TASK5

MILKA NAVEENA

TASK 5 — MODEL EVALUATION

5.1 Accuracy

```
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Accuracy (%) :", accuracy * 100)
```

Output

```
Accuracy: 0.XX
Accuracy (%) : XX.XX
```

```
Accuracy: 0.78
Accuracy (%) : 78.00
```

5.2 Precision

```
precision = precision_score(
    y_test,
    y_pred,
    zero_division=0
)

print("Precision:", precision)
```

Output

```
Precision: 0.XX
```

5.3 Recall

```
recall = recall_score(
    y_test,
    y_pred,
    zero_division=0
)

print("Recall:", recall)
```

Output

```
Recall: 0.XX
```

5.4 Classification Report

```
print(
    classification_report(
        y_test,
        y_pred,
        target_names=label_encoder.classes_,
        zero_division=0
    )
)
```

Output

	precision	recall	f1-score	support
Fail	0.xx	0.xx	0.xx	xx
Pass	0.xx	0.xx	0.xx	xx
accuracy			0.xx	79
macro avg	0.xx	0.xx	0.xx	79
weighted avg	0.xx	0.xx	0.xx	79

5.5 Confusion Matrix

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(7,5))
```

```
sns.heatmap(  
    cm,  
    annot=True,  
    fmt="d",  
    cmap="Blues",  
    xticklabels=label_encoder.classes_,  
    yticklabels=label_encoder.classes_  
)  
  
plt.title("Confusion Matrix")  
plt.xlabel("Predicted Label")  
plt.ylabel("Actual Label")
```

Output

		Predicted	
		Fail	Pass
Actual	Fail	TN	FP
	Pass	FN	TP

Report Explanation

The confusion matrix provides a detailed view of the model's predictions. It shows correctly classified students as well as students that were incorrectly classified. This helps identify whether the model is better at recognizing Pass or Fail cases.